

Route G-R8: A Closed Guided Clock Protocol and Its Complete Effective Phase

Same spectral action, explicit controls, no fitted time law

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Abstract

Specify a symmetric guided interferometer, its full scalar potential, preparation and analysis pulses. The ground propagation phase cancels for this particular potential, whereas the internal-energy gravitational phase survives. Both transport ramps are included. A finite-mass harmonic diagnostic quantifies, rather than assumes away, the excited packet's closure error. Pure/dephased controls and a partial-transpose witness separate entanglement from visibility. The result is a closed conservative effective protocol, not a demonstrated trapped-ion apparatus or an all-orders relativistic/open-system prediction.

1 Action and declared approximation

Keep the G-R4–G-R7 scalar thermal gaps at common local temperature T_0 :

$$E_i = h\nu_i(T_0), \quad \nu_i(T_0) = \nu_i^0[1 + b_i(T_0)], \quad H_C = \sum_{i=2,3} E_i |e_i\rangle\langle e_i|. \quad (1)$$

Each run prepares g and one selected e_i . These are not two independent qubits in one ion. The ground mass m absorbs any uniform ground-level shift; no missing ground polarizability is assigned a numerical value. Expand the same proper-time action in a weak uniform potential gz :

$$S_r = \int \left[-mc^2 + \frac{1}{2}m\dot{z}^2 - mgz - V_r + i\hbar\chi^\dagger\dot{\chi} - \chi^\dagger H_C \chi (1 + gz/c^2 - \dot{z}^2/(2c^2)) \right] dt, \quad (2)$$

$$H_r = mc^2 + \frac{p^2}{2m} + mgz + V_r + H_C \left(1 + \frac{gz}{c^2} - \frac{p^2}{2m^2c^2} \right). \quad (3)$$

The expansion keeps leading internal mass-energy coupling and Newtonian external dynamics. It does not retain all post-Newtonian center-of-mass terms. Metric, quantum mechanics and classical controls remain assumed. Guided-clock formalism and compensated transport have established precedents [1, 2]; the following control card is our explicit idealization, not an implementation imported from those papers.

There are two orthogonal guide modes A, B , initially/finally at the same height. A unitary coupler mixes them into two output ports; two orthogonal modes are never mapped into the same state. Other spatial degrees of freedom are matched and factorized. Temperatures are locally matched, not inferred from a global Tolman-equilibrium condition. The time-dependent guide can supply work: a closed path does not mean an energetically isolated atom. Controller backreaction and records are excluded only by the stated classical-control approximation.

2 Full path and full potential, not just a force

Let t_r be one ramp duration, t_h the hold and $t_f = 2t_r + t_h$. For $s(u) = 10u^3 - 15u^4 + 6u^5$, define the arm separation

$$q(t) = \begin{cases} d s(t/t_r), & 0 \leq t \leq t_r, \\ d, & t_r < t < t_r + t_h, \\ d[1 - s((t - t_r - t_h)/t_r)], & t_r + t_h \leq t \leq t_f, \end{cases} \quad q_A = q/2, \quad q_B = -q/2. \quad (4)$$

Position, velocity and acceleration close. Specify the scalar potential

$$V_r(z, t) = \frac{m\omega^2}{2}(z - q_r)^2 - m(\ddot{q}_r + g)z + w_r(t), \quad w_A = w_B = 0. \quad (5)$$

Its uniform support field cancels the ground mgz term. At zeroth order in $H_C/(mc^2)$,

$$\ddot{z} + \omega^2(z - q_r) = \ddot{q}_r. \quad (6)$$

Thus the packet center follows q_r . The accelerating-frame wave function is a rigid translated/boosted harmonic packet. For initial oscillator eigenstate n , its phase is

$$\psi_r(z, t) = e^{i[m\dot{q}_r(z - q_r) + S_r^c]/\hbar} e^{-i(\epsilon_n + mc^2)t/\hbar} \phi_n(z - q_r), \quad \dot{S}_r^c = \frac{1}{2}m\dot{q}_r^2 + m\ddot{q}_r q_r. \quad (7)$$

Direct substitution into the Schrödinger equation verifies this result. Since $[q_r \dot{q}_r]_0^{t_f} = 0$,

$$S_r^c = -\frac{m}{2} \int_0^{t_f} \dot{q}_r^2 dt, \quad S_A^c - S_B^c = 0. \quad (8)$$

The rest-energy and oscillator phases are common. At closure both packets agree, up to the explicitly computed equal phases.

A physically important nonuniqueness. Replacing the last z in Eq. (5) by $z - q_r$ changes no force but adds $m(\ddot{q}_r + g)q_r$ to the scalar potential. Its arm difference integrates to $mg\mathcal{A}$, where

$$\mathcal{A} = \int_0^{t_f} q(t) dt = d(t_h + t_r), \quad \int_0^{t_f} \dot{q}_r^2 dt = \frac{5d^2}{7t_r}. \quad (9)$$

The resulting ground fringe changes by $-mg\mathcal{A}/\hbar$. This is a different physical potential, not permission to discard a branch phase. A compensating transformation of coupler phases would be needed to interpret it as a mere basis convention. The specified $w_A - w_B = 0$ is a control requirement, not a consequence of force closure.

3 Pulses and the total observable phase

At common height prepare the clock with $R_y(\pi/2)$, with its optical phase defined by the local oscillator. Alternatively prepare its energy-dephased mixture with the same populations. Apply the path pulse

$$B(\alpha) = \exp[-i\pi(\cos \alpha X_P + \sin \alpha Y_P)/4] = \frac{I - i(\cos \alpha X_P + \sin \alpha Y_P)}{\sqrt{2}}. \quad (10)$$

Use $B(\alpha_s)$ at $t = 0$ and $B(\alpha_f)^\dagger$ after the full return. The card sets $\alpha_s = 0$ and scans α_f . These are impulse-limit operations, state-independent for the path coupler and with negligible differential vertical clock recoil. Finite pulses and real guide Stark shifts are not asserted to vanish in an apparatus.

Internal evolution follows directly from Eq. (2):

$$\theta_{ir} = 2\pi\nu_i(T_0) \int_0^{t_f} [1 + gq_r/c^2 - \dot{q}_r^2/(2c^2)] dt, \quad (11)$$

$$\delta_i = \theta_{iA} - \theta_{iB} = 2\pi\nu_i(T_0) \frac{g\mathcal{A}}{c^2}. \quad (12)$$

A finite packet's common momentum variance also cancels from this difference. The symmetric kinetic terms cancel; both ramps contribute dt_r to $\mathcal{A}$. During the stationary hold, $\Delta\nu_i = \nu_i(T_0)gd/c^2$, consistent with the phase derivative. For unequal baths one must instead integrate $2\pi\nu_i^0[(1+b_{iA})\lambda_A - (1+b_{iB})\lambda_B]$ and account for their forces and ground shifts. No such extra bath is silently introduced.

For $\rho_{gg} = \rho_{ee} = 1/2$, direct multiplication of both path pulses and the branch propagators gives

$$P(0, g) = \frac{1}{4}[1 + \cos \Phi_g], \quad \Phi_g = \alpha_f - \alpha_s, \quad (13)$$

$$P(0, e_i) = \frac{1}{4}[1 + \cos \Phi_{e_i}], \quad \Phi_{e_i} = \alpha_f - \alpha_s - \delta_i. \quad (14)$$

The other port uses the minus signs. Consequently

$$P(0) = \frac{1}{2}\{1 + \cos(\delta_i/2) \cos[\alpha_f - \alpha_s - \delta_i/2]\}, \quad \mathcal{V} = |\cos(\delta_i/2)|. \quad (15)$$

For this declared guide, the previously unresolved scalar phase is now calculated, not simply omitted. Adding actual scalar offsets changes both Φ_j by $-\int(w_A - w_B)dt/\hbar$. State-resolved phase differences cancel a stable common offset; uncontrolled shot-to-shot phase fluctuations can still destroy the signal.

All preparation/readout events occur at common height. Common clock pulse and free-precession phases do not enter Eq. (14), but must be calibrated for non-energy-basis joint measurements. There are no unspecified laser kicks at the separated positions in this protocol.

4 Mass-dependent closure: a finite Gaussian diagnostic

The same guide should not be declared exactly closing for every internal mass. Test it with $m_i = m(1 + \eta)$, $\eta = E_i/(mc^2)$ and

$$H_{r,i}^{\text{diag}} = m_i c^2 + p^2/(2m_i) + m_i g z + V_r. \quad (16)$$

This mass-only harmonic completion agrees with Eq. (3) to first order in η , but does not supply omitted relativistic terms. Write the actual center as $z_{A/B} = z_c \pm z_d/2$, with zero initial position and velocity. Then $\Omega = \omega/\sqrt{1+\eta}$ and

$$\ddot{z}_c + \Omega^2 z_c = -\frac{\eta}{1+\eta}g, \quad z_c = -\frac{\eta g}{\omega^2}(1 - \cos \Omega t), \quad (17)$$

$$z_d = q + y, \quad \ddot{y} + \Omega^2 y = -\frac{\eta}{1+\eta}\ddot{q}. \quad (18)$$

Define $F(\Omega) = \int_0^{t_f} \ddot{q}(t) e^{i\Omega t} dt$. The return acceleration is minus the outbound one, shifted by $t_r + t_h$:

$$F(\Omega) = \frac{d}{dt_r} \int_0^1 s''(u) e^{i\Omega t_r u} du [1 - e^{i\Omega(t_r+t_h)}]. \quad (19)$$

Let $I_c + iI_s = e^{i\Omega t_f} F(\Omega)^*$. At final closure of q ,

$$z_d(t_f) = -\frac{\eta I_s}{(1+\eta)\Omega}, \quad \dot{z}_d(t_f) = -\frac{\eta I_c}{1+\eta}, \quad \int_0^{t_f} z_d dt = \mathcal{A} + \frac{\eta I_c}{\omega^2}. \quad (20)$$

Choose both ωt_r and ωt_h to be integer multiples of 2π . Then $F(\omega) = 0$: first-order mass-dependent displacement closes. Finite η residuals are nonzero, not an exact all-mass theorem.

The exact phase of the two Gaussian packets' overlap includes the endpoint term $-m_i \dot{z}_c z_d / \hbar$. The classical action difference is $m_i z_c \dot{z}_d - \eta m g \int z_d dt$. Combining them yields

$$\arg\langle \psi_{B,i} | \psi_{A,i} \rangle = -\delta_i - \frac{m\eta^2 g}{\hbar\omega^2} \Re F(\Omega). \quad (21)$$

Thus closing central positions alone would have missed a phase term. For the initial ground Gaussian with covariance $V_0 = \text{diag}(\hbar/(2m\omega), \hbar m\omega/2)$, set

$$V = S_i V_0 S_i^T, \quad S_i = \begin{pmatrix} \cos \Omega t_f & \sin \Omega t_f / (m_i \Omega) \\ -m_i \Omega \sin \Omega t_f & \cos \Omega t_f \end{pmatrix}. \quad (22)$$

Its overlap modulus is $r_i = e^{-D_i}$, where

$$D_i = \frac{\Delta p^2 V_{zz} + \Delta z^2 V_{pp} - 2\Delta z \Delta p V_{zp}}{2\hbar^2}, \quad \Delta z = z_d(t_f), \quad \Delta p = m_i \dot{z}_d(t_f). \quad (23)$$

The energy-resolved excited fringe gets factor r_i , and the unconditioned clock overlap becomes $(1 + r_i e^{-i\delta_i^{\text{eff}}})/2$. This is a computed motional correction, not an invented environmental rate.

5 Pure/dephased comparison and a small-phase witness

After known local phase changes, the ideal pure joint state is

$$|\Psi_\delta\rangle = e^{i\delta Z_P Z_C/4} |++\rangle = \cos(\delta/4) |++\rangle + i \sin(\delta/4) |--\rangle. \quad (24)$$

For example the diagonal internal evolution factors as $e^{-i\bar{\theta}P_e}e^{-i\delta Z_P/4}e^{i\delta Z_P Z_C/4}$, where $\bar{\theta} = (\theta_A + \theta_B)/2$. The initial splitter's $-i$ phase is another local path rotation. These phase changes require calibrated controls, not a claim that a large optical phase is known from arbitrary decimal precision.

Replacing $|+\rangle\langle+|_C$ by $I_C/2$ leaves all energy-resolved fringes and the reduced path state unchanged. The resulting joint state is a convex sum of product states conditioned on g, e and is separable. For the pure state, $\mathcal{N} = |\sin(\delta/2)|/2$. For the positive small phase in our card, use

$$W = \frac{1}{4}(I - X_P X_C + Z_P Y_C + Y_P Z_C). \quad (25)$$

Indeed $W = (|v_-\rangle\langle v_-|)^{T_C}$ with $|v_-\rangle = (|+-\rangle - i|++\rangle)/\sqrt{2}$ in the X basis. For any separable ρ , $\rho^{T_C} \geq 0$, proving $\text{tr}(W\rho) \geq 0$. Direct evaluation gives

$$\langle W \rangle_{\text{pure}} = -\frac{1}{2} \sin(\delta/2), \quad \langle W \rangle_{\text{dephased}} = \frac{1}{4}[1 - \sin(\delta/2)]. \quad (26)$$

For the opposite phase sign, reverse the signs of the two Y terms. This is an operational witness, not an inference from visibility.

Actual measurement settings. Measure XX, ZY, YZ in separate runs. For either subsystem, analysis $R_y(-\pi/2)$ followed by population (Z) detection measures X ; $R_x(\pi/2)$ followed by Z measures Y ; no pulse measures Z . The three witness operators commute globally, despite requiring incompatible local axes. To explicitly include noncommuting joint controls, also measure ZZ :

$$[Z_P Y_C, Z_P Z_C] = 2iI_P X_C \neq 0. \quad (27)$$

Together with independently characterized coherent/dephased preparation, these settings go beyond energy-basis populations. Finite-mass Gaussian corrections, noise and pulse errors must be included in a measured state; the ideal witness values are not observed violations.

Statistical cost should not be hidden. For the ideal pure state $\langle XX \rangle = 1$ and $\langle ZY \rangle = \langle YZ \rangle = -\sin(\delta/2)$. With N independent shots for each setting,

$$\sigma_W^2 = \frac{\cos^2(\delta/2)}{8N}, \quad N > \frac{9 \cos^2(\delta/2)}{8\mathcal{N}^2} \quad (28)$$

is the ideal three-standard-error separation criterion. This is not a rigorous finite-sample confidence bound or an experimental sensitivity: it excludes state-preparation/readout errors, drift, losses and count rate.

6 Numerical result, rejection criteria and next decision

The single illustrative card fixes $d = 1$ mm, $t_r = 5$ ms, $t_h = 10$ ms, $\omega/(2\pi) = 1$ kHz, $g = 9.80665$ m/s², $T_0 = 300$ K and a rounded $m = 171$ u . The latter is not a precision ionic-mass determination. It gives $t_f = 20$ ms, $\mathcal{A} = 1.5 \cdot 10^{-5}$ m s and $\Delta\tau = 1.63670545 \cdot 10^{-21}$ s. Each arm reaches speed 0.1875 m/s and acceleration 115.47 m/s²; neither establishes hardware feasibility.

Quantity	E2	E3
Hold $\Delta\nu$ (Hz)	$7.51094 \cdot 10^{-5}$	$7.00642 \cdot 10^{-5}$
Full δ (rad)	$7.07889 \cdot 10^{-6}$	$6.60339 \cdot 10^{-6}$
Ideal $1 - \mathcal{V}$	$6.26384 \cdot 10^{-12}$	$5.45060 \cdot 10^{-12}$
Ideal $\mathcal{N} = -\langle W \rangle$	$1.76972 \cdot 10^{-6}$	$1.65085 \cdot 10^{-6}$
Ideal shots per witness setting	$3.59205 \cdot 10^{11}$	$4.12798 \cdot 10^{11}$

The effective total phases are $\Phi_g = \alpha_f - \alpha_s$ and $\Phi_e = \Phi_g - \delta$, not just a hold-only clock phase. Both pure and dephased initial clocks have the same path visibility. The latter has $\langle W \rangle \simeq 0.25$, not a negative witness.

The harmonic diagnostic gives $|\Delta v| \simeq 7.0 \cdot 10^{-23}$ m/s (E2), $1 - r \simeq 5.2 \cdot 10^{-40}$ and a phase correction of order 10^{-30} rad. These tiny numbers are cancellation checks in a perfect model, not metrological claims. High precision evaluates the known small- η cancellation; it does not fit any parameter. Ordinary trap errors will dominate. The modified centered potential of section2 would instead add roughly $-3.96 \cdot 10^5$ rad of common ground phase while leaving its forces unchanged.

Useful physical gates. An actual state-dependent guide energy $u_{i,r}(t)P_{e_i}$ adds $\int (u_{i,A} - u_{i,B})dt/\hbar$ to δ_i and is degenerate with the desired effect unless independently calibrated. For this card, keeping that bias below the signal requires a time-averaged differential frequency bias below roughly $5.6 \cdot 10^{-5}$ Hz (E2) or $5.3 \cdot 10^{-5}$ Hz (E3), before demanding any precision margin. Uncalibrated guide scalar offsets, controller which-path records, finite pulse recoil, lifetime and radiation noise remain physical limitations. The existing tens-of-ms E2 lifetime caution still applies to a 20 ms sequence; no survival rate or complete dissipator is imported.

Next decision: choose and calibrate one real guide/coupler and its state-resolved phase noise, or reject this implementation if the requirements are not credible. Prioritize differential phase; pursue the joint witness only with an adequate statistics/control budget. Do not respond to poor feasibility by fitting a new time law, claiming exact all-mass closure, or enlarging the derivation indefinitely.

References

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